25CS101 PROGRAMMING IN C

Hours Per Week :

L	T	P	SL	C
2	0	4	2	4

PREREQUISITE KNOWLEDGE: Fundamentals of Problem Solving.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed to impart knowledge on basic concepts of C programming language and problem solving through programming. It covers basic structure of C program, data types, operators, decision making statements, loops, functions, strings, pointers, structures and unions. At the end of this course, students will be able to design, implement, test and debug complex problems using features of C.

MODULE-1

UNIT-1 **12L+0T+24P +12SL=48 Hours**

INTRODUCTION TO ALGORITHMS AND PROGRAMMING LANGUAGES:

Introduction to Algorithms: Introduction to computational thinking – Problem solving with algorithms and flowcharts - Introduction to C: Structure of a C program - pre-processor statement) inline comments, variable declaration statements, executable statements; C Tokens - C character set, identifiers and keywords, type qualifiers, type modifiers, variables, constants, punctuations and operators.

Data Types and Operators: Basic data types; Formatted I/O; Reading and writing characters; Operators - assignment, arithmetic, relational, logical, bitwise, ternary, address, indirection, sizeof, dot, arrow, parentheses operators; Expressions - operator precedence, associative rules.

Control Statements: Introduction to category of control statements; Conditional branching statements - if, if– else, nested-if, if – else ladder, switch case;

Iterative statements - for, while, do - while, nested loops; Jump statements - break, jump, goto and continue.

Algorithmic logic in Panini's Ashtadhyayi; problem-solving in Indian logic ***(Nyaya Shastra)***, Structured numerical systems – Vedic mathematics and early binary concepts

UNIT-2

FUNCTIONS & POINTERS:

User-defined functions: Function declaration - definition, header of a function, body of a function, function invocation; Call by value, Call by address. Recursion; Library Functions; Storage classes; Scope of a variable.

Pointers: Declaration, Initialization, Multiple indirection, Pointer arithmetic.

Recursive hymns ***(Rigveda),*** modular constructs in Shastra classification, Concepts of abstraction and references in Indian epistemology ***(Pramana).***

MODULE-2

UNIT-1 **18L+0T+36P +18SL=72 Hours**

ARRAYS:

Arrays: Introduction; Types of arrays; Single dimensional array - declaration, initialization, usage, reading, writing, accessing, memory representation, operations; Multidimensional arrays. Passing arrays to functions; Relationship between arrays and pointers, scaling up - array of arrays, array of pointers, pointer to a pointer and pointer to an array; Dynamic memory allocation functions – Creating arrays dynamically and performing operations on them.

Representation of multidimensional data in ancient Indian astronomy and calendar systems Granularity and scalability in system design found in traditional architecture (e.g., ***Vastu***).



Image Source: https://m.media-amazon.com/images/I/61V713nRFeL._SL1280_.jpg

SKILLS:

- ✓ *Analysis of the problem to be solved.*
- ✓ *Select static or dynamic data structures for a given problem and manipulation of data items.*
- ✓ *Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt.*

UNIT-2

STRINGS:

Strings: Character array, reading strings from the standard input device, displaying strings on the standard output device, Importance of terminating a string, Standard string library functions.

Preservation and transmission of Sanskrit and Tamil texts using structured phonetic systems.

UNIT-3

STRUCTURES, UNIONS:

Structures: Defining a structure, declaring structure variable, Operations on structures, Pointers to structure - declaring pointer to a structure, accessing structure members using pointer; Array of structures, Nested structures, passing structures to functions - passing each member of a structure as a separate argument, passing structure variable by value, passing structure variable by reference/ address; Typedef and structures.

Unions: Defining a union - declaring union variable, operations on union.

Classification of knowledge in Indian systems – Charaka Samhita, Manuscripts, Palm-leaf layouts, Group representation in indigenous knowledge frameworks – e.g., Ayurveda Tridosha classification.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1	Explain algorithmic problem-solving, flowchart design, and fundamental programming constructs in C.	Apply	1, 2
2	Analyze and implement various control structures (conditional, iterative, and jump statements) for effective decision-making in programs.	Analyze	2, 3, 4
3	Implement modular programming by designing user-defined functions and utilize recursion effectively for problem-solving. Apply array concepts, including single and multi-dimensional arrays, dynamic memory allocation, and pointer-array relationships in programming.	Apply	1, 2, 3, 4, 5
4	Utilize string handling techniques and standard string library functions to solve text-processing problems effectively.	Evaluate	1, 2, 3, 5
5	Design, implement, and analyze structured data types using structures and unions for efficient handling of complex data.	Create	1, 2, 3, 5

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) AND INDIAN KNOWLEDGE SYSTEM (IKS):



TEXTBOOKS:

1. Behrouz A. Forouzan, Richard F.Gilberg, "Programming for Problem Solving", 1st edition, Cengage publications, 2019.
2. Ajay Mittal, "Programming in C - A Practical Approach", 1st edition, Pearson Education, India, 2010.

REFERENCE BOOKS:

1. 1. R.G. Dromey, "How to Solve it by Computer", Prentice-Hall International Series in Computer Science, USA.
2. Reema Thareja, "Computer Fundamentals and Programming in C", 1st edition, Oxford University Press, India, 2013.
3. Herbert Schildt, "C: The Complete Reference", 4th edition, Tata McGraw-Hill, 2017.
4. Byron S Gottfried, "Programming with C", 4th edition, Tata McGraw-Hill, 2018.